

2(a)	(resultant) force proportional to rate of change of momentum	B1
2(b)(i)	arrow drawn vertically downwards from point X	B1
2(b)(ii)	$s = ut + \frac{1}{2}at^2$ $h = \frac{1}{2} \times 9.81 \times 0.81^2$	C1
	= 3.2 m	A1
2(b)(iii)	$d = 5.4 \times 0.81$ = 4.4 m	A1
2(c)(i)	downward pointing arrow labelled weight	B1
	upward pointing arrow labelled air resistance	B1
2(c)(ii)	air resistance increases	B1
	weight constant or <u>resultant</u> force decreases	B1
	(so) acceleration decreases	B1
2(c)(iii)	gravitational potential energy to thermal/internal energy	B1